INCH-POUND

MIL-PRF-12574M
9 June 1997
SUPERSEDING
MIL-R-0012574L (GL)
22 May 1992
MIL-R-12574K
30 June 1988

PERFORMANCE SPECIFICATION

REFRIGERATION UNITS, MECHANICAL, PANEL TYPE, 5000 BTU FOR REFRIGERATOR, FIELD, PORTABLE (150 CUBIC FOOT)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers two types of panel type, mechanical, refrigeration units having a capacity of 5000 British thermal units (Btu) per hour and used with a 150-cubic foot, portable field refrigerator (see 6.1).

1.2 Classification. The refrigeration units will be of the following types as specified (see 6.2):

Type I - Diesel Engine Driven
Type II - Electric Motor Driven

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5018 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.2 Government documents .

2.2.1 Standards . The following standards form part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-461	- Requirements for the Control of Electromagnetic Interference Emissions and Susceptibility
MIL-STD-462	- Measurement of Electromagnetic Interference Characteristics
MIL-STD-810	- Environmental Test Methods and Engineering Guidelines
MIL-STD-1474	- Noise Limits for Military Materiel (Metric)

(Unless otherwise indicated, copies of the above standards are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094).

2.3 Non-Government publications . The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

Standards Publication No. 250 - Enclosures for Electrical Equipment

(Application for copies should be addressed to the National Electrical Manufacturers Association, 1300 No. 17th Street, Suite 1847, Rosslyn, VA 22209).

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

Standard No. 70 - National Electrical Code

(Application for copies should be addressed to the National

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Fire Protection Association, One Battery March Park, Quincy, MA 02269-9101).

UNDERWRITERS LABORATORIES INC. (UL)

Standard 207- Refrigerant Containing Components and Accessories, Nonelectrical
Standard 486- Wire Connectors and Solder Lugs

(Application for copies should be addressed to the Underwriters Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062-2096).

2.4 Order of precedence . In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article . When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.1.1.

3.2 **Operating requirements**. Refrigeration units shall satisfy the following functional, operational and performance requirements.

3.2.1 4-hour operation . Refrigeration units shall operate properly with suction and discharge pressures safely controlled.

3.2.2 72-hour operation . Refrigeration units shall operate properly and maintain noted conditions at thermostat settings of 0^{0}F and 35 ^{0}F.

3.2.3 Refrigeration capacity . Refrigeration units shall have minimum net capacities of 5,000 Btu/hr at a thermostat setting of 0^{0}F and 7,500 Btu/hr at a thermostat setting of 35 ^{0}F.

3.2.4 Defrost system . Refrigeration units shall have a defrost cycle terminated by temperature, automatically initiated by a time control adjustable between 6 and 24 hours and have the capacity to produce not less than 6 pounds of condensate within 30 minutes.

3.2.5 Moisture content . The moisture content of the refrigeration system shall be kept at a minimum to maintain proper operation of the refrigeration unit.

3.2.6 Refrigerant leakage . Refrigerant containing components shall be gastight.

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3.2.7 Operation time . Type I refrigeration units shall operate for a minimum of 38 hours without refueling.

3.2.8 Control panel . The refrigeration units shall have a control panel for the operator having, but not limited to, the following items:

1. Start, run, stop control
2. Operation indicator
3. Refrigerator temperature control
4. Refrigeration unit running time indicator
5. High and low side pressure indicators
6. Refrigerator interior temperature indicator
7. Defrost operation indicator
8. Fuel capacity indicator for type I units
9. Battery charging circuit voltage indicator for type I units

3.2.9 Ozone-depleting substances . The refrigerant shall be commercially available nonozone-depleting, and environmentally safe. Class I ozone-depleting substances, as determined by the Environmental Protection Agency, shall not be contained in, and shall be prohibited from use during the fabrication of the refrigeration units (see 6.6).

3.2.10 Temperature control . The selected temperature of the refrigeration units shall be controlled automatically. The temperature control shall allow an incremental selection of normal refrigerator temperatures.

3.2.11 Diesel engine operation . Unless otherwise specified for type I refrigeration units (see 6.2), the diesel engine's electrical system shall operate on 12-volt direct current (DC). The engine shall operate on diesel and JP-8 fuels.

3.2.12 Power requirements . Unless otherwise specified for type II refrigeration units (see 6.2), the refrigeration units shall operate on 208/230 volt, 3-phase, 60-hertz (Hz) and 3-phase, 50-Hz power on the voltage specified in the contract or order (see 6.2).

3.2.13 Finish . The refrigeration units except where detrimental to proper operation shall have a nontoxic chemical agent resistant green finish (see 6.7.1). The word "CARC", together with the month and year of finishing, shall be stencilled near the identification plate in block letters not less than 1 inch high using lusterless black chemical agent resistant paint (see 6.7.2) All paint shall be properly applied.

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3.2.14 Safety. The refrigeration units shall minimize flammability, operational, toxic and personnel hazards.

3.2.15 Airflow. The air exiting the evaporator shall be directed to create proper circulation inside the refrigerator it is used with.

3.3 **Support or ownership requirements**. Refrigeration units shall meet the following requirements.

3.3.1 Weight. The net weight of the refrigeration units, including oil and refrigerant, shall be not more than 775 pounds for type I units, excluding the battery and fuel, and not more than 775 pounds for type II units.

3.3.2 Transportability. Refrigeration units shall be able to be handled by a forklift truck.

3.3.3 Design reliability. Refrigeration units shall operate properly for not less than 2500 hours.

3.3.4 Service and maintenance. All major components shall be accessible for adjustment, maintenance, servicing, and replacement without removing any other major component. All other components and assemblies shall be designed for ease of installation and replacement and shall be accessible for maintenance, repair, service and replacement with minimum disturbance to adjacent components. No special tools shall be needed for any maintenance, repair, service, installation or replacement action. Condenser and evaporator fins shall be protected from any damage that might occur during servicing or maintenance actions. Operator personnel shall be able to visually check the moisture content of the refrigeration system on all refrigeration units and the compressor crankcase oil level on type I units.

3.3.5 Marking. Identification, instruction, control and gage plates shall be of corrosion-resistant material permanently attached to the outside of the refrigeration unit. Numbering and lettering shall be legible and permanently inscribed and shall be no less than 1/16 inch in height. All plates shall blend with the framed enclosure finish and shall have a nonreflective surface.

3.3.5.1 Identification plate. The identification plate shall contain the following information:

1. U.S.

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2. Military model number and nomenclature
3. National Stock Number (NSN)
4. Serial number
5. Contract number
6. Manufacturer's name and address
7. Manufacturer's model number
8. Commercial and Government Entity (CAGE) number
9. Technical manual number, if applicable, (see 6.2 and 6.5
10. Weight
11. Refrigerant used

3.3.5.2 Instruction plates . Instruction plates shall contain the wiring and refrigeration cycle diagrams, instructions for start-up, operation, shutdown and preventive maintenance.

3.3.5.2.1 Electrical instruction plate . Type II refrigeration units shall have an electrical instruction plate conforming to all requirements of 3.3.5, except it shall be located near the electrical supply junction box. The plate shall be visible from outside the refrigeration unit and shall contain the wording "For 60-hertz operation use 208/230 volt, 3-phase power source only. For 50-hertz operation use "x" volt, 3-phase power source only." The voltage specified in the contract or order (see 6.2) for 50-hertz operation shall be marked in the "x" location.

3.3.5.3 Control and gage nameplates . All controls and gages shall have identifying nameplates.

3.3.5.4 Electrical wiring . Electrical wiring shall be marked to allow identification with the wiring diagram.

3.3.6 Interchangeability . All component parts of the refrigeration unit shall be interchangeable with the same parts of other units furnished under the same contract.

3.4 **Environmental requirements**. Refrigeration units shall operate under the following environmental conditions.

3.4.1 Ambient temperature . The refrigeration units shall operate in ambient temperatures from 0 ^{0}F to 125 ^{0}F.

3.4.2 Rainfall . The refrigeration unit shall operate in rainfall.

3.4.3 Shock and vibration . The refrigeration units shall be operable after being subjected to shock and vibration.

3.4.4 Electromagnetic compatibility . When specified (see 6.2), the type I and type II refrigeration units shall not exceed radiated emission (RE)102 limits for ground applications (Navy fixed and AF) over a range of 2 megahertz (MHz) to 400 MHz. When

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specified (see 6.2), type II refrigeration units shall not exceed conducted emission (CE)102 limits.

3.4.5 Storage conditions . The refrigeration units shall remain operable after storage in temperatures of minus 65 ^{0}F to plus 155 ^{0}F.

3.4.6 Noise level . The noise levels at personnel "operator's" and "servicing" positions shall not damage personnel hearing.

3.4.7 Corrosion resistance . The refrigeration units shall operate after exposure to a salt fog atmosphere.

3.4.8 Grounding for shipboard use . When type II units are procured for Navy shipboard use (see 6.2), provisions shall be made to permanently ground and bond all interior and exterior metal sheathing and the frame of the motor to a grounding rod.

3.5 **Interface and interoperability requirements** . The refrigeration units shall be designed for use with 150-cubic foot refrigerators (see 6.1) and shall have overall dimensions no greater than those shown on figure 1.

4. VERIFICATION

4.1 Classification of inspections . The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.1.1)
b. Conformance inspection (see 4.1.2)

4.1.1 First article inspection . When a first article inspection is required (see 3.1 and 6.2), a sample shall be examined for all the verifications listed in table I.

TABLE I. Verification methods

Title	Require-ment	Verification	Title	Require-ment	Verification

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OPERATING REQUIREMENTS	3.2	4.3.2	Design reliability	3.3.3	4.3.3.3
4-hour operation	3.2.1	4.3.2.1	Service and maintenance	3.3.4	4.3.3.4
72-hour operation	3.2.2	4.3.2.2	Marking	3.3.5	4.3.3.5
Refrigeration capacity	3.2.3	4.3.2.3	Identification plate	3.3.5.1	4.3.3.5.1
Defrost system	3.2.4	4.3.2.4	Instruction plates	3.3.5.2	4.3.3.5.2
Moisture content	3.2.5	4.3.2.1	Electrical instruction plate	3.3.5.2.1	4.3.3.5.2.1
Refrigerant leakage	3.2.6	4.3.2.1	Control and gage nameplates	3.3.5.3	4.3.3.5.3
Operation time (type I)	3.2.7	4.3.2.2d	Electrical wiring	3.3.5.4	4.3.3.5.4
Control panel	3.2.8	4.3.2.5	Interchangeability	3.3.6	4.3.3.6
Ozone-depleting substances	3.2.9	4.3.2.6	**ENVIRONMENTAL REQUIREMENTS**	3.4	4.3.4
Temperature control	3.2.10	4.3.2.7	Ambient temperature	3.4.1	4.3.4.1
Diesel engine operation (type I)	3.2.11	4.3.2.8	Rainfall	3.4.2	4.3.4.2
Power requirements (type II)	3.2.12	4.3.2.9	Shock and vibration	3.4.3	4.3.4.3
Finish	3.2.13	4.3.2.10	Electromagnetic compatibility	3.4.4	4.3.4.4
Safety	3.2.14	4.3.2.11	Storage conditions	3.4.5	4.3.4.5
Airflow	3.2.15	4.3.2.1	Noise level	3.4.6	4.3.4.6
SUPPORT OR OWNERSHIP REQUIREMENTS	3.3	4.3.3	Corrosion resistance	3.4.7	4.3.4.7
Weight	3.3.1	4.3.3.1	Grounding for Navy	3.4.8	4.3.4.8
Transportability	3.3.2	4.3.3.2	**INTERFACE AND INTEROPERABILITY REQUIREMENTS**	3.5	4.3.5

4.1.2 Conformance inspection . Conformance inspection of the refrigeration units shall consist of all the verifications listed in table I, except the verification testing specified for the requirements of paragraphs 3.2.2 through 3.2.4, 3.2.7, 3.3.2, 3.3.3 and 3.4.1 through 3.4.7 shall not be done. Sampling for this

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inspection shall be as specified in the contract or order (see 6.2).

4.2 Order of verifications . Verification listed in table I can be performed in any order except verification tests shall be conducted in the order shown in table II.

TABLE II Verification test sequence

Test sequence	Requirement Paragraph	Characteristic	Verification Paragraph
1	3.2.1	4-hour operation	4.3.2.1
2	3.2.2	72-hour operation	4.3.2.2
3	3.2.3	Refrigeration capacity	4.3.2.3
4	3.2.4	Defrost system	4.3.2.4
5	3.4.1	Ambient temperature (high)	4.3.4.1.1
6	3.4.3	Shock and vibration	4.3.4.3
7	3.3.2	Transportability	4.3.3.2
8	3.4.2	Rainfall	4.3.4.2
9	3.4.1	Ambient temperature (low)	4.3.4.1.2
10	3.4.4	Electromagnetic compatibility	4.3.4.4
11	3.4.5	Storage conditions	4.3.4.5
12	3.4.6	Noise level	4.3.4.6
13	3.4.7	Corrosion resistance	4.3.4.7
14	3.3.3	Design reliability	4.3.3.3

4.3 Verification methods . Verification methods can include visual examination, measurement, testing, simulation, modeling, engineering evaluation, component properties analysis, certification and similarity to previously-approved or previously-qualified designs.

4.3.1 Verification alternatives . The contractor may propose alternative test methods, techniques, or equipment, including the

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application of statistical process control, tool control, or cost-effective sampling procedures to verify performance (see 6.4).

4.3.2 **Operating requirements verification**. All operating requirements shall be verified as follows.

4.3.2.1 4-hour operation test . The refrigeration unit evaporator shall be clamped to the evaporator opening of a calorimeter or other insulated structure. All refrigeration unit components shall be in place and the refrigeration unit doors shall be closed. The refrigeration unit shall be operated for 4 hours to determine that all mechanical and electrical components are functioning as intended and all controls are in calibration. The units shall have a heat load applied to the enclosure so that the unit operates more than 50 percent of the test time and cycles on the automatic controls three or more times per hour. Suction pressure at the compressor shall be verified to be within the established guidelines of the compressor manufacturer. Prior to completion of the 4-hour test, the air flow to the condenser shall be blocked off and the unit operated and automatic shutdown of the refrigeration unit verified when compressor discharge limit is exceeded. During normal operation, the discharge pressure shall be observed to be not less than that recommended by the compressor manufacturer for the refrigerant used. The air exiting the evaporator shall be checked to verify that the airflow direction is out and upward.

After a minimum operating time of 1 hour with the thermostat set at 0 ^{0}F, the moisture indicator shall be observed to verify that the moisture content as shown by the component manufacturer's color chart data is not greater than 25 parts per million. A check for refrigerant leakage shall also be performed with the unit inoperative with a leak detector having a maximum sensitivity of 0.40 oz per year. Leakage shall be checked to verify that it is no greater than 0.5 oz per year, except the compressor shaft seal leakage is no greater than 3.0 oz per year.

4.3.2.2 72-hour operating test . A 72-hour test shall be conducted as follows:

a. The refrigeration unit shall be clamped to the evaporator opening of a calorimeter or other insulated structure.

b. A combined heat load of 3000 Btu/hr shall be impressed when the thermostat is set at 0 ^{0}F and 4000 Btu/hr when the thermostat is at the 35 ^{0}F setting.

c. The refrigeration unit shall be operated 36 hours with the thermostat set at 0 ^{0}F and 36 hours with the thermostat set at 35^{0}F. During this test, suction and discharge pressure, ambient air, condenser air, and evaporator in and out air temperatures shall be recorded on automatic recording instruments. Normal

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maintenance shall be performed during this test and verification shall be made that rated conditions are maintained during the 72-hour period in an ambient temperature of 110 ^{0}F.

d. When the refrigeration unit being tested is a type I unit, the fuel consumption shall be monitored and recorded. Verification shall be made that the unit operates no less than 38 hours before refueling.

4.3.2.3 Refrigeration capacity tests .

4.3.2.3.1 Calorimeter calibration . The calorimeter calibration shall be run once prior to capacity tests to establish the coefficient of heat leakage to be used for capacity calculation. The coefficient of heat leakage for the calorimeter shall not exceed 25 Btu/hr/ ^{0}F and the volume shall not be less than 150 cubic feet. A refrigeration unit shall be bolted to the evaporator opening in the calorimeter with a minimum 5-foot clearance on the front of the unit and the overall heat leakage coefficient shall be determined as follows:

a. An electric fan(s), of sufficient capacity to insure that no uncirculated air pocket will exist, shall be installed inside the calorimeter. The electrical energy of the fan motor shall be measured by the same meter used for the heating elements in subparagraph (c) and this heat equivalent shall be added to other sources of heat added to the calorimeter.

b. Four thermocouples, shielded from the effects of radiant heat, shall be installed inside the calorimeter and positioned within 6 inches of the ceiling and floor at two opposite corners. The interior temperature shall be taken as the average of the four thermocouples.

c. Heating elements shall be installed approximately in the center of the calorimeter. Electrical heat input shall be measured by a watt-hour meter with a scale increment of 0.1 kilowatt-hour or less.

d. Eight thermometers or thermocouples, shielded from the effects of the radiant heat, shall be installed approximately 18 inches from each corner of the calorimeter in the room where it is placed. The ambient temperature shall be taken as the average of these eight temperatures.

e. All openings in the calorimeter shall be sealed, and the calorimeter shall remain undisturbed throughout the test.

f. The ambient temperature shall then be maintained constant within $\pm$ 2^{0}F. Sufficient, steady, non-varying heat shall be added

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to the interior of the calorimeter to maintain a minimum of 60 ^{0}F and a maximum of 100 ^{0}F temperature difference between the interior and exterior of the calorimeter.

g. The test shall be considered concluded when eight consecutive half-hourly readings indicate that:

1. The ambient temperature has been maintained constant within $\pm 2^0$F.
2. The temperature inside the calorimeter has been maintained constant within $\pm 1^0$F.

h. The overall heat leakage coefficient U (Btu/hr/ ^{0}F) of the calorimeter is given by:

$$U = \frac{H_1}{(T_1 - T_2)}$$

H_1 = Sum of all electrical heat energies added to the calorimeter, Btu/hr.

T_1 = Average interior temperature of the calorimeter, (100 ^{0}F to 155 ^{0}F).

T_2 = Average ambient temperature, ($T_1 - T_2$) shall be in the range of 60 ^{0}F to 100 ^{0}F.

i. The test calorimeter shall not be used unless a heat leakage coefficient no greater than 25 Btu/hr/ ^{0}F is obtained.

4.3.2.3.2 Capacity test (0 ^{0}F). The capacity of the refrigeration unit for low temperature shall be verified to be no less than 5,000 Btu/hr using the method described below. Recorded data shall be required and shall include all specified temperature readings, electrical heat input, suction and discharge pressures, and complete calculations. All panels and doors of the refrigeration unit shall be in the closed position.

a. All accessories and appurtenances shall be in their proper place.

b. Thermocouples, shielded from the effects of radiant heat, shall be installed in the air streams entering the evaporator and leaving the evaporator section.

c. Shielded thermocouples shall be located at the approximate center of four equal areas of the condenser face. The condenser inlet temperature will be taken as the average of these four temperatures.

d. Recording pressure gages shall be used for the compressor suction and discharge pressures.

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e. The procedures described in 4.3.2.3.1 (b), (c), (d), and (e) shall be followed for this test.

f. The refrigeration unit shall then be operated with air surrounding the condenser section and entering the condenser at a minimum of 110 ^{0}F and sufficient, steady, non-varying heat shall be added to the interior of the calorimeter to maintain 0 ^{0}F temperature of air entering the evaporator, $\pm$ 1^{0}F. A watt-hour meter with scale increments of 0.1 kilowatt-hour or less shall be used to indicate the quantity of electrical energy added. Air velocity shall not exceed 100 feet per minute at any point 3 feet from the refrigeration unit.

g. The test shall be considered concluded when eight consecutive half-hourly readings indicate that:

1. The average calorimeter ambient temperature, T_3 has been maintained at steady state of $\pm$ 2^{0}F at a temperature of 100 ^{0}F to 110^{0}F.

2. Return air temperature, T_4, entering the evaporator has been maintained at 0 ^{0} F $\pm$ 1^{0}F.

3. The net refrigerator capacity shall be considered as the sum of the electrical energy, in Btu/hr, added during this test plus the heat leakage gain in Btu/hr, as calculated from the calibration specified in 4.3.2.3.1.

The net refrigeration capacity is given by:

$Q = U (T_3 - T_4) + H_2$
Q = Net refrigeration capacity in Btu/hr
U = Overall heat leakage coefficient Btu/hr/ ^{0}F as determined by paragraph (h)in 4.3.2.3.1
H_2 = Sum of all heat energy added to the interior of the calorimeter in Btu/hr, (f) above.

4.3.2.3.3 Capacity test (35 ^{0}F). The procedure specified in 4.3.2.3.2 shall be followed, with the following exceptions, to verify that the capacity of the refrigeration unit is not less than 7,500 Btu/hr.

a. In 4.3.2.3.2 (b), the thermocouples shall be installed as specified. However, the temperature difference between the air entering the evaporator and leaving the evaporator section shall not be recorded.

b. In 4.3.2.3.2 (f), sufficient heat shall be added to maintain 35^{0}F $\pm$ 1^{0}F temperature of air entering the evaporator in place of 0 ^{0}F temperature specified.

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c. In 4.3.2.3.2 (g) 2, temperature of the air entering the evaporator has been maintained at 35 ^{0}F $\pm$ 1^{0}F in place of 0 ^{0}F temperature specified.

4.3.2.4 Defrost test . With the refrigeration unit operating in a maximum ambient temperature of 20 ^{0}F and the insulated structure maintained at 0 ^{0}F + 10^{0}F, the evaporator shall be frosted up by spraying water on the coil surface or evaporating water in the box. After the coil is frosted, the refrigeration unit shall be set on the defrost cycle and the condensate collected and weighed. Verification shall be made that the unit operates through a complete defrost cycle, that no less than 6 pounds of condensate is produced in 30 minutes, that no overloading of the compressor motor or diesel engine occurs, that there is no damage to the compressor because of lack of lubrication or refrigerant "slugging" and that no condensate gets into the food storage area when the refrigeration cycle begins again. Verify that there is no ice on the coil at the end of the defrost cycle.

4.3.2.5 Control panel inspection . During the 4-hour operation test, verification shall be made that the on/off switch and operation indicator works properly; the refrigeration temperature control and internal refrigerator temperature indicator agree; pressure reading indicators agree with pressures being recorded, and the operation running time indicator agrees with the test timing device. For type I units the fuel capacity and voltage indicators shall be visually checked for proper operation. Functioning of the defrost operation indicator shall be visually verified during the defrost test.

4.3.2.6 Ozone-depleting chemicals inspection . Visually check publication SD-14 (see 6.6) to verify that the refrigerant used in the refrigeration unit is not listed.

4.3.2.7 Temperature control inspection . Visually inspect the temperature control to verify that temperatures from minus 10 ^{0}F to plus 50 ^{0}F can be selected in increments not less than 2 ^{0}F and not greater than 6 ^{0}F.

4.3.2.8 Diesel engine operation demonstration . During the 72-hour operation test on type I refrigeration units, observe that diesel fuel is used for one 36-hour period of operation and JP-8 fuel is used for the other 36-hour period of operation.

4.3.2.9 Power requirements demonstration . During the 72-hour operation test on type II refrigeration units, observe that the unit operates on a 208/230 volt, 3-phase, 60-Hz power

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source for one 36-hour period of operation and on 3-phase, 50-Hz power with voltage as specified in the contract for the other 36-hour period.

4.3.2.10 Finish certification and examination . The contractor shall certify that the refrigeration unit's finish is nontoxic and chemical agent resistant, meeting government requirements. Painted surfaces shall be visually examined to verify that there are no streaks, crops, pinholes, checks or cracks.

4.3.2.11 Safety certification and inspection . Safety requirements shall be verified as follows.

4.3.2.11.1 Wiring certification . Wiring practices used in the refrigeration units shall be certified as conforming to NFPA Standard No. 70 by submission of the manufacturer's certification, a certified test report or appropriate labels or listing marks.

4.3.2.11.2. Electrical components and enclosure certification . Electrical components shall be certified as conforming to the requirements of UL Standard 486 and electrical enclosures (junction boxes, control boxes, etc.) shall be certified as conforming to the watertightness requirements of NEMA Standards Publication No. 250 by submission of the manufacturer's certification, a certified test report or appropriate labels or listing marks.

4.3.2.11.3 Refrigeration components certification . Refrigerant containing components used in the refrigeration units shall be certified as conforming to UL Standard 207 by submission of the manufacturer's certification, a certified test report or appropriate labels or listing marks.

4.3.2.11.4 Toxic hazards inspection . Visually inspect type I refrigeration units to verify that provisions for oil drainage and discharge of exhaust gases are located outside the refrigeration unit.

4.3.2.11.5 Personnel safety inspection . Visually verify that a plate indicating "Danger, Rotating Fan" has been permanently attached on both sides of the condenser section when applicable.

4.3.3 **Support or ownership requirements verification** . All ownership and support requirements shall be verified as follows.

4.3.3.1 Weight inspection . Verify, by weighing, that the type I unit weighs no more than 775 pounds, excluding battery and fuel, and

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that the type II unit weighs no more than 775 pounds.

4.3.3.2 Transportability test . The refrigeration unit loaded to 1 1/2 times the weight of the unit, shall be lifted, transported, and placed in position by a forklift truck without securing the unit in any way to the truck. During the transport, the truck will be driven over a 4-inch by 4-inch board so either the two left or two right tires go over the board. Upon completion, visually examine the refrigeration unit to verify that there is no damage or distortion.

4.3.3.3 Design reliability test . A 2500-hour design reliability test shall be run as follows:

a. A maximum of two units shall be used for this test. When two units are used, each unit shall be tested for a minimum of 1000 operating hours and their combined testing time must equal 2500 operating hours. Operating hours for the units shall be the hours the compressor is operating.

b. The refrigeration unit shall be clamped to the evaporator opening of the calorimeter (see 4.3.2.3.1) with a minimum volume of 150 cubic feet.

c. A combined heat load of approximately 4000 Btu/hr shall be imposed on the unit so the compressor will cycle with running time of approximately 75 percent when the condenser air entering temperature is 110 ^{0}F.

d. The air entering the condenser and surrounding the condensing section shall be maintained at a minimum of 110 ^{0}F for 90 percent of the operating hours, and at a minimum of 125 ^{0}F for 10 percent of the operating hours.

e. The defrost control shall be set for two defrosts in a 24-hour period of operating time.

f. The refrigeration unit shall be operated with the thermostat set at 0 ^{0} F for 36 hours and at 40 ^{0} F for the next 36 hours. This cycle shall be repeated until all operating hours have been completed. During the test, suction and discharge pressures, air temperature entering the condenser, refrigerated enclosure temperature, and compressor operating hours shall be recorded every hour on automated recording instruments.

g. Failure of the unit to maintain the temperature of the enclosure, failure to start and operate, failure to defrost, or any

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mechanical or control failure that requires more than 30 minutes to repair, shall be defined as a unit failure. A malfunction that can be repaired in less than 30 minutes shall be classified as an incident.

h. When failure to complete test hours occurs, the failure shall be analyzed, correction made, and the test started from time zero. All failures, analyses, and corrective actions shall be recorded.

i. The test shall be successfully completed when the unit has completed all operating hours without failure and with no more than five incidents.

4.3.3.4 Service and maintenance simulation . A simulated demonstration shall be accomplished verifying that all major refrigeration unit components are accessible for adjustment, maintenance, servicing and replacement without the removal of any other major component or need for special tools. The simulation shall also illustrate that condenser and evaporator fins have guards or other protection and all other components and assemblies are accessible for maintenance, repair, service or replacement and require no special tools for those actions. Verification shall be made that the moisture content of the refrigeration system and the compressor crankcase oil level (type I units) can be visually determined.

4.3.3.4.1 Alternative service and maintenance analysis . When a technical manual is furnished as part of the contract, it can be analyzed for proper service and maintenance procedures in lieu of the simulation specified in 4.3.3.4.

4.3.3.5 Marking inspection . Visually verify that all identification, instruction, control and gage plates are permanently attached to the outside of the refrigeration unit, lettering is permanently inscribed, legible and not less than 1/16 inch high, and plates have a nonreflective surface.

4.3.3.5.1 Identification plate inspection . Visually verify that all information is complete and accurate.

4.3.3.5.2 Instruction plates inspection . Visually verify that all diagrams and instructions are complete and accurate.

4.3.3.5.2.1 Electrical instruction plate inspection . Visually verify that type II refrigeration units have an electrical instruction plate containing correct power source information located near the electric supply junction box.

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4.3.3.5.3 Control and gage nameplate inspection . Visually verify that all controls and gages have nameplates.

4.3.3.5.4 Electrical wiring inspection . Visually verify that all wiring is marked to allow identification with the wiring diagram on the instruction plates.

4.3.3.6 Interchangeability certification . The contractor shall certify that all component parts of the refrigeration unit shall be interchangeable with the same parts of other units of the same type furnished under the same contract.

4.3.4 **Environmental requirements verification** . All environmental requirements shall be verified as follows.

4.3.4.1 Ambient temperature testing . All refrigeration units shall be tested for operation in high temperature ambients as specified in 4.3.4.1.1. Type I refrigeration units shall also be tested for operation in low temperature ambients as specified in 4.3.4.1.2.

4.3.4.1.1 High temperature operation test . The refrigeration unit shall be heat soaked for 6 hours at an ambient temperature of 125 ^{0}F minimum. With air entering and surrounding the condenser and the condenser section maintained at 125 ^{0}F minimum, the unit shall be started and operated for 1 hour with its control set at 40 ^{0}F, then shut off. Air velocity around the condenser section shall not exceed 100 feet per minute. After 2 minutes, the unit shall be restarted and operated for one hour with its control set at 40 ^{0}F, then shut off. This procedure shall be followed for a total of 4 hours with the unit continuing to operate.

4.3.4.1.2 Low temperature operation test . The type I units, complete with battery and fuel, shall be placed in a temperature controlled room maintained at 0 ^{0}F, $\pm$ 2^{0}F, for 12 hours. On completion of the cold soak period, the unit shall be automatically started and operated for a period of 1 hour during which time the cold room shall be maintained at 0 ^{0}F $\pm$ 15^{0}F.

4.3.4.2 Rainfall test . The refrigeration unit shall be mounted in a simulated container and fresh water sprayed on the unit at no less than 3 inches per hour at an average downward angle of 45 degrees to simulate rainfall on the exposed surfaces of the unit. The front, top, and two sides of the unit shall be individually sprayed for 30 minutes while not operating with the spray applied perpendicular to the surfaces. The refrigeration unit shall then be started and operated for 30 minutes while the spray continues. A visual verification shall be done to insure no leakage occurred in

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the evaporator section, electrical control or junction boxes, the food storage area of the simulated container, or the muffler/exhaust system (type I).

4.3.4.3 Shock and vibration test . While operating, the refrigeration unit shall be raised 1 inch above the floor and dropped onto a concrete surface 50 times. The unit shall continue to be operated for 2 hours, including one defrost cycle. The unit shall be checked to insure that there is no damage, fuel leakage (type I), refrigerant leaks, distortion or malfunction of any component.

4.3.4.4 Electromagnetic compatibility tests . When electromagnetic compatibility is required, the refrigeration unit shall be tested in accordance with MIL-STD-462. An EMI test report shall be compiled documenting the results of all tests performed. Type I refrigeration units shall meet the RE102 limits of MIL-STD-461. Type II units shall meet the RE102 and CE102 limits of MIL-STD-461.

4.3.4.5 Storage test . The refrigeration unit shall be subjected to a temperature of 155 ^{0}F for 4 hours, then the room temperature shall be brought to 70 ^{0}F $\pm$ 5^{0}F and the unit operated for 1 hour. The refrigeration unit shall then be subjected to a temperature of -65 ^{0}F for 4 hours, then the room temperature shall be increased to 70 ^{0}F $\pm$ 5^{0}F and the unit operated for 1 hour. Verify that there has been no permanent set or loss of resiliency of packing or gaskets, congealing of lubricants, discoloration, cracking, bulging, checking or crazing.

4.3.4.6 Noise limit test . The refrigeration unit shall be tested for noise level as specified in MIL-STD-1474. The noise limit category shall be "D". The "operator's" position shall be 10 feet in front of the unit and 5 feet above the ground level. The "servicing" position shall be 2 feet in front of the control panel and 4 feet above the ground level. The noise level shall be less than 85 decibels.

4.3.4.7 Corrosion resistance test . The refrigeration unit shall be tested for its resistance to the effects of a salt fog atmosphere having a concentration of 5 percent $\pm$ 1 percent. The unit, in its operational configuration, shall be tested in accordance with MIL-STD-810, Method 509.3, with the refrigeration unit subjected to 24 hours of salt fog exposure followed by 24 hours of drying time in an ambient temperature of 70 ^{0}F $\pm$ 5^{0}F, followed by a second 24-hour exposure and 24-hour drying time. The unit shall then be operated and visually checked over all surfaces to verify that there is no corrosion.

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4.3.4.8 Grounding for Navy inspection . Visually verify that type II units procured for Navy shipboard use incorporate grounding provisions.

4.3.5 **Interface and interoperability requirements measurement** . The refrigeration units shall be measured to determine that overall dimensions are no greater than indicated on figure 1.

5. PACKAGING

5.1 Packaging . For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory).

6.1 Intended use . The refrigeration units covered by this specification are intended for worldwide field use by the Military under adverse climatic conditions without added protection from the elements. The refrigeration unit is used with the 150-cubic foot refrigerators covered by MIL-R-12571.

6.2 Acquisition requirements . Acquisition documents must specify the following:

a. Title, number, and date of this specification
b. Type of refrigeration unit required (see 1.2)
c. Issue of DoDISS to be cited in the solicitation, and, if required, the specific issue of individual documents referenced (see 2.2.1 and 2.3)
d. When a first article is required (see 3.1, 4.1.1 and 6.3)
e. When the voltage requirement is other than specified (see 3.2.11)
f. Voltage requirement when operation on a 3-phase, 50-Hz power source is required (see 3.2.12 and 3.3.5.2.1)
g. Voltage, phase, and cycle requirements when the power supply is other than specified see 3.2.12)

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h. Technical manual number, when required (see 3.3.5.1)
i. When electromagnetic compatibility is required (see 3.4.4 and 4.3.4.4)
j. When procurement is for Navy (see 3.4.8)
k. Sampling plan for conformance inspection (see 4.1.2)
l. Packaging requirements (see 5.1)

6.3 First article . When requiring a first article inspection, contracting documents should provide specific guidance to offerors. This guidance should cover whether the first article is a first article sample, a first production item, or the number of test items. These documents should also include specific instructions regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Pre-solicitation documents should provide government waiver rights for samples for first article inspection to bidders offering a previously-acquired or tested product. Bidders offering such products who wish to rely on such production testing must furnish evidence with the bid that prior Government approval is appropriate for the pending contract.

6.4 Verification alternatives . Contracting documents should provide guidance to offerors regarding the submission of alternatives to specified verification methods (see 4.3.1).

6.5 Technical manuals . The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, specifications and standards that have been cleared and listed in DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements list (DD Form 1423), which is included as an exhibit to the contract.

The technical manuals must be acquired under a separate contract line item in the contract.

6.6 Ozone-depleting substances . A listing of ozone-depleting substances prohibited from use can be found in publication SD-14 Listing of Toxic Chemicals, Hazardous Substances, and Ozone-Depleting Chemicals. It is available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.

6.7 Finish .

6.7.1 Paint, green . Treatment and painting in accordance with MIL-C-53072 or types F or G of MIL-T-704 have been used to satisfy this requirement (see 3.2.13).

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6.7.2 Paint, black . A paint in accordance with MIL-C-53039 or types III or IV of MIL-C-46168 have been used to satisfy this requirement (see 3.2.13).

6.8 Recycled, recovered, or environmentally preferable materials . Recycled, recovered or environmentally preferable materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements and promotes economically advantageous life cycle costs.

6.9 Subject term (key word) listing .

Diesel-powered
Electric-powered
Food preservation, field
Food storage, field
Refrigeration, field

6.10 Supersession data . MIL-R-0012574L (GL), which provides a detailed configuration of the refrigeration unit, is superseded by this performance specification. MIL-R-0012574L (GL) and associated drawings are available to the contractor on an information basis.

6.11 Changes from previous issue . Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:

Army - GL
Navy - YD1
Air Force - 99

Preparing activity:

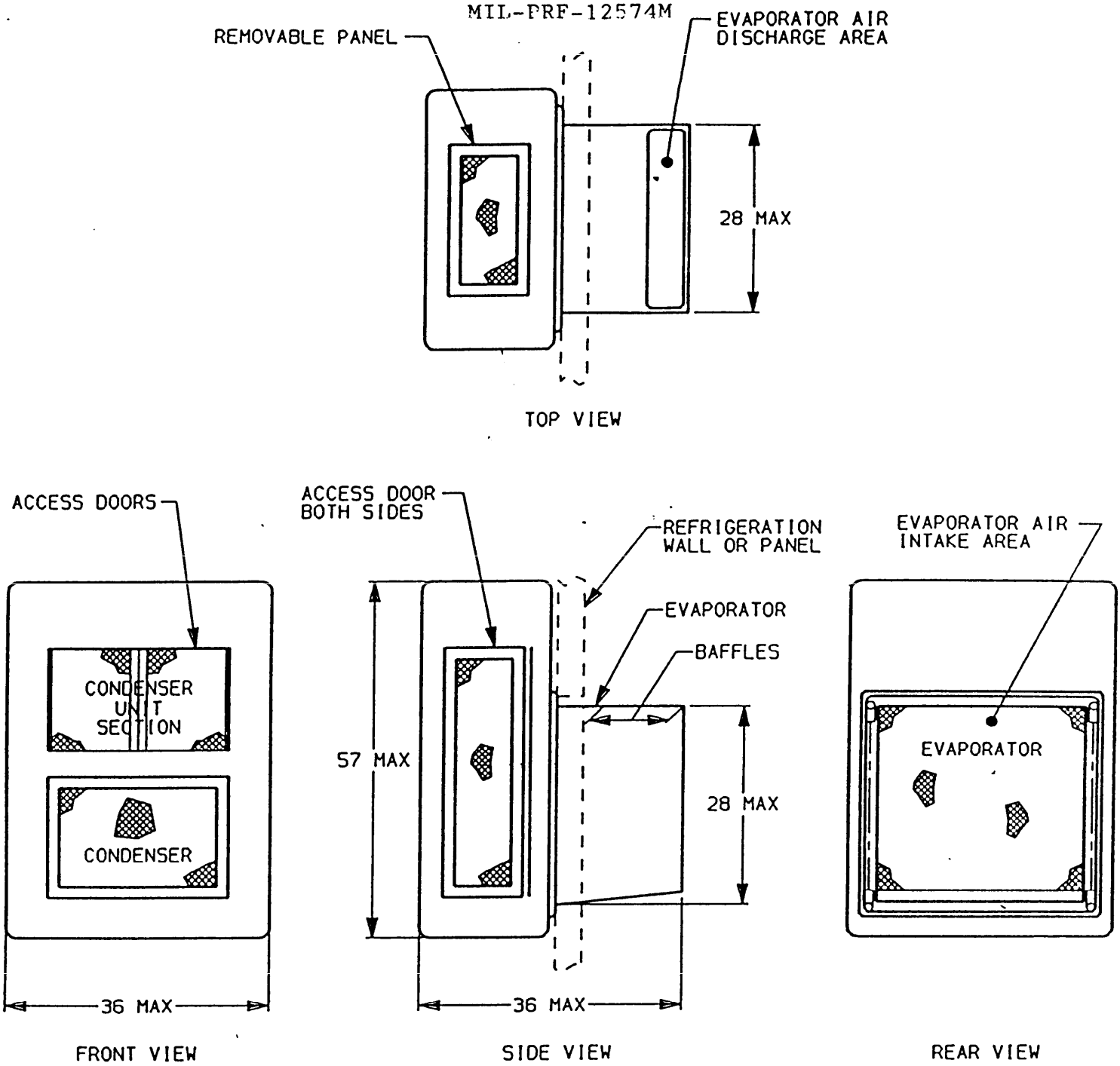
Army - GL

Project 4110-0500)

Review activities:

Army - AV, CE, MD1, ME, SM
Navy - MC
Air Force - 82
DLA - IS

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NOTES:

1. DETAILS OTHER THAN SPECIFIC DIMENSIONS ARE SHOWN FOR INFORMATION PURPOSES ONLY.
2. EVAPORATOR HEIGHT AND WIDTH DIMENSIONS ARE OVERALL DIMENSIONS AND MUST BE HELD TO PERMIT EVAPORATOR TO FIT IN A 28 1/2 X 28 1/2 OPENING.
3. MAXIMUM WIDTH, DEPTH AND HEIGHT DIMENSIONS INCLUDE ALL PROTRUSIONS.
4. UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN INCHES.

FIGURE 1
DIMENSIONAL REQUIREMENTS FOR 5000 BTU REFRIGERATION UNIT

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